



Large Language Models for Tabular Data: Progresses and Future Directions

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ABSTRACT

Tables contain a significant portion of the world's structured information. The ability to efficiently and accurately understand, process, reason about, analyze, and generate tabular data is critical for achieving Artificial General Intelligence (AGI) systems.

However, despite their prevalence and importance, tables present unique challenges due to their structured nature and the diverse semantics embedded within them. Textual content, numerical values, visual formats, and even formulas in tables carry rich semantic information that is often underutilized due to the complexity of accurately interpreting and integrating.

Fortunately, the advent of Large Language Models (LLMs) has opened new frontiers in natural language processing (NLP) and machine learning (ML), showing remarkable success in understanding and generating text, code, etc. Applying these advanced models to the domain of tabular data holds the promise of significant breakthroughs in how we process and leverage structured information.

Therefore, this tutorial aims to provide a comprehensive study of the advances, challenges, and opportunities in leveraging cutting-edge LLMs for tabular data. By introducing methods of prompting or training cutting-edge LLMs for table interpreting, processing, reasoning, analytics, and generation, we aim to equip researchers and practitioners with the knowledge and tools needed to unlock the full potential of LLMs for tabular data in their domains.

CCS CONCEPTS

• **Information systems** → **Information retrieval**.

KEYWORDS

Tabular data, Large language models, Representation learning

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1 DETAILED SCHEDULE

The tutorial, as an in-person event, adopts the length of three hours plus breaks. A detailed schedule is as follows:

- **PART I.** Introduction to This Tutorial (20 minutes)
 - Motivation (10 minutes)
 - Scope (i.e. breadth/depth) (10 minutes)
- **PART II.** Table Representation for LLMs (35 minutes)
 - Fundamental elements of tabular data (5 minutes)
 - Representation approaches of tabular data (20 minutes)
 - Retrieval or compression methods for large tables or databases (10 minutes)
- **PART III.** Training LLMs for Tabular Data (25 minutes)
 - Corpora of tabular data (5 minutes)
 - Continued pre-training (10 minutes)
 - Single-task & multi-task fine-tuning (10 minutes)
 - Question and Answer Session (10 minutes)
 - Coffee Break —
- **PART IV.** Tasks and Benchmarks (20 minutes)
 - Table interpreting and processing (10 mins)
 - Table reasoning and analysis (10 mins)
- **PART V.** LLM-driven Table Agents (40 minutes)
 - Tabular environments (10 minutes)
 - Agents: decompose tasks, use tools, sampling and refine (30 minutes)
- **PART VI.** Hands-on Session and New horizons (15 minutes)
 - Demonstration & Hands-on session (10 mins)
 - New horizons (5 minutes)
 - Question and Answer Session (10 minutes)

2 RELEVANCE

This tutorial holds strong relevance to the core domains of SIGIR where the topics of table data retrieval, machine learning, and data intelligence are of paramount interest. This tutorial will not only build upon the knowledge shared in previous tutorials such as "Web Table Extraction, Retrieval and Augmentation" [67] at SIGIR 2019 and "From Tables to Knowledge: Recent Advances in Table Understanding" [47] at KDD 2021, but will also introduce the latest developments in LLMs that were not covered previously. Recent surveys on LMs for tabular data include [14, 15, 21, 41, 70].

3 BRIEF CONTENT

3.1 Structured, Semi-structured, and Image Tables

Well-structured tables, such as those found in databases, feature a clearly defined relational schema. This format facilitates precise data retrieval using formal languages like SQL, R, and Python's

Pandas [15]. In contrast, semi-structured tables, common in web pages, LaTeX documents, spreadsheets, and Word documents, often lack necessary meta-information to accurately capture the table’s high-level structure and data types, posing challenges for precise and automatic information retrieval [17, 35]. Additionally, a substantial volume of image tables exists, containing only raw RGB data [14], and PDF tables are typically treated as such [49].

3.2 Table Representation for LLMs

The process of converting tables into prompts while preserving their semantic integrity presents the first challenge. LLMs necessitate prompts in a linear, sequential text format, which contrasts with the inherently two-dimensional structure of tables that possess a defined schema. The prompt might encompass its content, schema, visual formats, foreign keys, formulas, etc.

A prevalent method to encode table content involves linearizing the tables row by row, employing column separators such as “|” to demarcate individual cells. Studies have assessed LLMs’ effectiveness across a spectrum of table representations, including CSV, markdown, JSON, XML, and HTML [50, 51].

Studies found that incorporating details like foreign keys and minimizing explanations can notably improve LLM performance on these tasks. To establish database connections, DBCopilot [53] streamlines the text-to-SQL conversion process by segmenting it into schema routing and SQL generation phases.

Visual formats play a vital role in interpreting tables, and studies exist on LLMs’ use of image-based table representations [14].

3.3 Retrieval or Compression Methods for Large Tables or Databases

Even LLMs with extensive contextual capabilities may fall short when dealing with exceedingly large tables. Furthermore, [6] discovered that GPT’s effectiveness dwindles to mere conjecture with tables exceeding 1,000 tokens in size. Additionally, longer prompts lead to increased costs, particularly for applications relying on LLM APIs.

To bridge this gap, [28, 39] suggested simple truncation strategies, limiting the input to a maximum sequence length. [51] implemented predefined constraints to align with LLM call requirements. Retrieval-augmented LLMs employ a retriever-reader framework to streamline the analysis of vast table collections by focusing on a select subset. Notable efforts [37, 38] showcase the segmentation of large tables for efficient retriever-reader training, enhancing performance. DB-GPT [60] provides a comprehensive solution that includes retrieval-enhanced generation and agent interactions, catering to industrial needs. [53] advances database querying by segmenting the process into schema understanding and SQL generation, thereby offering improved accuracy and scalability for handling extensive databases across domains.

3.4 Training LLMs for Tabular Data

Before the appearance of LLMs, researchers have introduced various Tabular Language Models (TaLMs) as summarized by [15]. These models modify LMs’ structures, encoding, and objectives to align with table-centric tasks. And recent works on LLM for tabular data commonly choose not maintain the model structure.

For multi-task fine-tuning, instruction tuning transcends the limitations of fine-tuning on singular datasets by embracing the fine-tuning of language models across a spectrum of datasets. This approach significantly bolsters performance on novel tasks and enhances model flexibility [58]. Table-GPT, for instance, employs a synthesis-followed-by-augmentation strategy for dataset creation, thereby facilitating a deep understanding of tabular data structures and relationships [36]. Projects like TableLlama [68] prefer real-world data from established datasets to synthesized data, showcasing the generalized applicability of LLMs post-instruction tuning. TableLLM¹, a 13B LLM, is designed for tabular data tasks in documents and spreadsheets, suitable for Office environments. The training employs a distant supervision approach, incorporating a reasoning process extension and cross-validation to ensure data quality. The Tabular LLM also aims to gather datasets for tasks like table QA and text generation, converting them into instruction-tuning formats² to enhance LLMs’ understanding of tabular data.

Multi-modal training is essential for handling image-based tables or leveraging visual formats like human-beings. While [14] provides insights into the strengths and limitations of LLMs in processing multimodal tabular data representations, this area remains underexplored and holds significant potential for future research.

3.5 Table Tasks and Benchmarks

Table interpreting and processing. Starting with the most basic process of table collection, many earlier works collect spreadsheet [10, 19] and HTML [13, 34] tables from the web. Relatedly, the tasks of table detection and classification are essential to building a table corpus of high quality. Table datasets with table type annotations such as FUSE [2], ENRON [26], EUSES [23], and DECO [32] can enable better table recognition systems. Besides understanding, Dong et al. [16] and Kardas et al. [31] also perform structure extraction for tables from spreadsheet files and scientific papers.

More granular than the table-level processing tasks, many tasks target detection and classification tasks on the header (region) [20] and cell levels, the latter of which is more widely explored by WebSheet [16], DeEx [32], SAUS [24], and CIUS [24]. In addition, other tasks are oriented towards knowledge in tables or concerning web data, such as column type prediction [4, 5], entity linking [3], relation identification [18, 22], table retrieval [27, 56, 66].

Table reasoning and analysis. Table reasoning tasks are mainly about verifying facts in and answering questions about tables. For fact verification, available datasets such as FreebaseQA [30] and TabFact [8] enable more accurate methods to detect factual errors [1, 25, 33, 48, 65, 72]. For table-based question answering, the WikiTableQuestions dataset [44] contains compositional questions about wiki tables, and OTT-QA [7] further extends to questions about both table and text; Spider [63] focus on database queries, HiTab [11] focus on complex hierarchical spreadsheets tables, and TabMWP [40] with math world problems ground on tables. Lastly, the data-to-text generation task assesses models in table analysis, such as ToTTo [43] on web tables and HiTab [11] on spreadsheets.

¹<https://tablellm.github.io/>

²https://github.com/PhoebusSi/Alpaca-CoT/tree/tabular_llm

3.6 LLM-driven Table Agents

LLMs face challenges with table-related tasks, such as arithmetic operations [11, 44] and factual lookups [9], due to their inability to interpret compositional queries in structured table environments. To address these limitations, LLM-powered agents emerge as a solution, capable of exerting actions on tables via various means.

Environment. Tabular information poses numerous challenges depending on the data source and the presented interface. As shown by the tasks discussed above, tables could come from various sources hence bringing along individual challenges: Wikipedia tables with semi-structured format [45], spreadsheet files with complex, hierarchical table structures [11], and even large databases that may exceed model input limit [63]. Further, depending on the table and method used, a table can be presented in different ways, such as a flattened text sequences, a structure-preserving pandas dataframe, or even a clickable UI interface supporting font and color changes. These interaction modes can be realized using certain agents with certain actions, which we introduce below.

Agent. Existing table agents are designed with techniques to decompose complex tasks, leverage external tools [55, 64], and perform sampling and iterative refinement [42]. Drawing inspiration from methods such as Chain-of-Thought (CoT) [59] and least-to-most prompting, researchers have guided LLMs to break down intricate table tasks into simpler, manageable sub-tasks [46, 57]. Similarly, for handling voluminous web tables, Dater [62] utilize LLMs to divide extensive data into smaller, more digestible pieces, then solve individual pieces to finally solve the complex query.

However, agents may not solve the tasks perfectly in one try. Therefore, research initiatives such as ReAct [61] propose an iterative, closed-loop system for table agents, which repeatedly analyze table data and user intentions, generate and execute action plans, and update their action trajectories until the desired outcome is achieved. To tackle multiple tasks at a time, StructGPT [29], adeptly addresses both table QA and text-to-SQL tasks by employing versatile actions suitable for web tables, databases, and knowledge graphs. Further, TAP4LLM [52] extends the utility of LLM agents to table augmentation tasks, showcasing the adaptability of LLM agents in more table domains. Another direction for agents to explore more solutions is sampling multiple generations and selecting the best via self-consistency [54], based on the assumption that the most common output is more likely to be correct. Nonetheless, the approach is shown effective in many works [35, 46, 62, 71].

Actions. Actions are essential to the performance of agents, which should be functionally sufficient to solve the table task, while not too difficult for the agent to acquire. Actions can be categorized into agent-inherent actions and application-specific tools. Besides the most atomic actions such as add, different works introduce application-specific actions. For example, SheetCopilot [35] add spreadsheet APIs as actions, Binder [12] use a knowledge-query action for web tables, Data-Copilot [69] add human-designed Python functions for individual downstream domains. TroVE [55] use LLMs to automatically build macro actions that suits the target task.

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